R2.2 How fast? The rate of chemical change

Maxwell-Boltzmann distribution curves

Learning outcomes

By the end of this section you should be able to:

- (R2.2.4) Sketch Maxwell-Boltzmann energy distribution curves to explain the effect of temperature on the probability of successful collisions.
- (R2.2.5) Sketch and explain energy profiles with and without catalyst for endothermic and exothermic reactions.
- (R2.2.5) Construct Maxwell-Boltzmann distribution curves to explain the effect of different values of E_a on the probability of successful collisions.

Activation energy, E_a

When you turn on the methane gas in your stove, it does not start burning as soon as you turn the knob. You need to ignite the gas for the reaction to start, once that is done the reaction proceeds spontaneously. The precise amount of energy needed to ignite the gas is called the activation energy (E_a) for the combustion of methane.

You have seen energy profile diagrams in section R1.1.3 (https://app.kognity.com/:sectionlink:138726). For an endothermic reaction the products are at a higher energy than the reactants as shown in **Figure 1**.

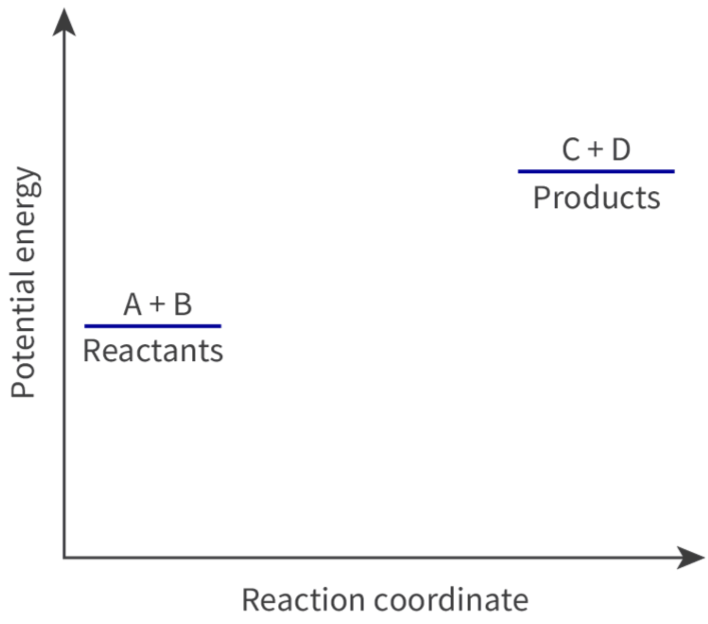

Figure 1. An energy profile diagram for an endothermic reaction.

More information for figure 1

This is an energy profile diagram for an endothermic reaction. The X-axis represents the reaction coordinate, while the Y-axis represents potential energy. The diagram shows two horizontal lines. The first line, labeled "A + B Reactants," is positioned lower on the Y-axis, indicating a lower potential energy level. The second line, labeled "C + D Products," is positioned higher on the Y-axis, indicating a higher potential energy level than the reactants. This illustrates the typical energy increase seen in an endothermic reaction, where the products have higher energy than the reactants.

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If we draw an energy profile diagram to show what actually happens to the energy during the reaction it would appear like this.

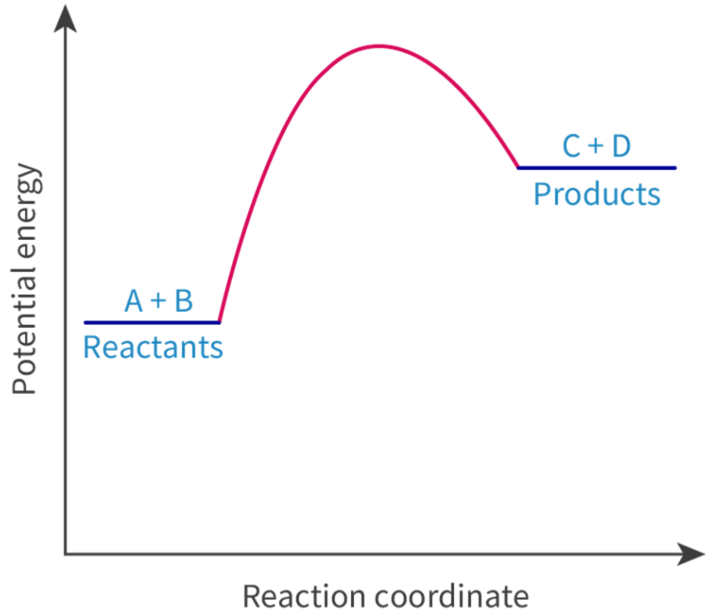

Figure 2. An energy profile diagram for an endothermic reaction showing the bump for activation energy.

More information for figure 2

The image is a diagram representing an energy profile for an endothermic reaction. It is a graph plotting potential energy on the vertical axis against the reaction coordinate on the horizontal axis. The diagram illustrates a curve starting from a lower energy level indicating the reactants labeled as 'A + B'. Then, it rises to form a peak, which represents the activation energy — the threshold energy required for the reaction to occur. After the peak, the curve descends to another energy level representing the products labeled as 'C + D'. The peak is higher than both the starting and ending points, showing that the reaction requires input energy to reach the transition state before forming products, which signifies an endothermic process. The axes are marked with 'Potential Energy' on the vertical and 'Reaction Coordinate' on the horizontal.

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What do you think the hump in the energy represents? The activation energy of a reaction is the minimum amount of energy required by reactant particles to form product particles. We know that for a collision to be successful it must have sufficient kinetic energy. This energy goes into breaking the bonds in the reactants so that the reaction may proceed. When this energy is available the reactants achieve a transition state from which they can then form products. So only the particles that have energy greater than or equal to the activation energy will undergo successful collisions. Other particles will still collide, but the collisions will not lead to a reaction.

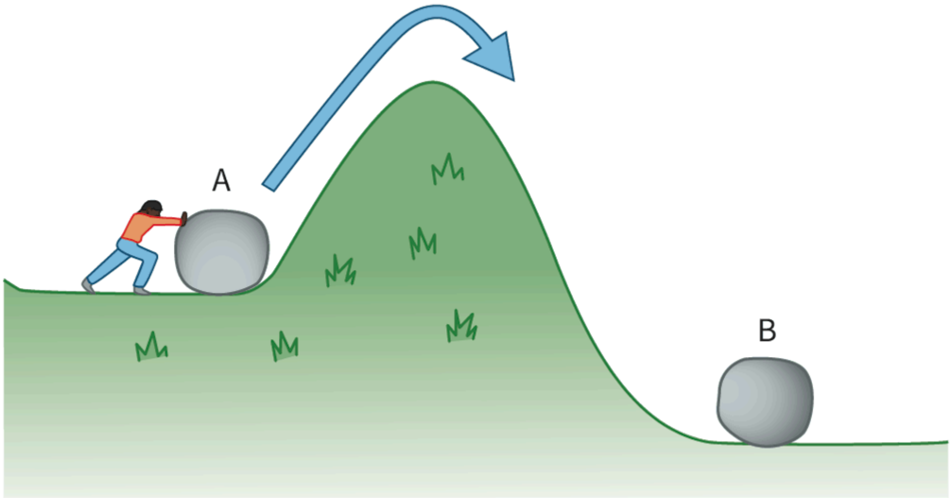

Figure 3. The activation energy barrier.

More information for figure 3

The image is a diagram depicting a person labeled 'A' pushing a rock up a grassy hill with a hump. The hill has a green surface with patches of grass. The person is positioned at the lower left side, exerting force on the rock to move it upward over the hump. An arrow is drawn over the hill, indicating the direction of movement from left to right, going over the hump and down towards another rock labeled 'B' at the bottom right of the hill. This illustrates the concept of an activation energy barrier in a chemical reaction where an initial push is needed to overcome a resistance before achieving a transition state.

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Concept

A transition state is a species that forms when a reaction has the activation energy needed to break the bonds in the reactants. The transition state is formed midway between the reactants and products, with partially broken and partially formed bonds. This makes the transition state unstable, and it quickly rearranges itself to form the products.

The energy profile diagrams in **Figure 4** show the activation energy for an endothermic and exothermic reaction. The transition state is also shown.

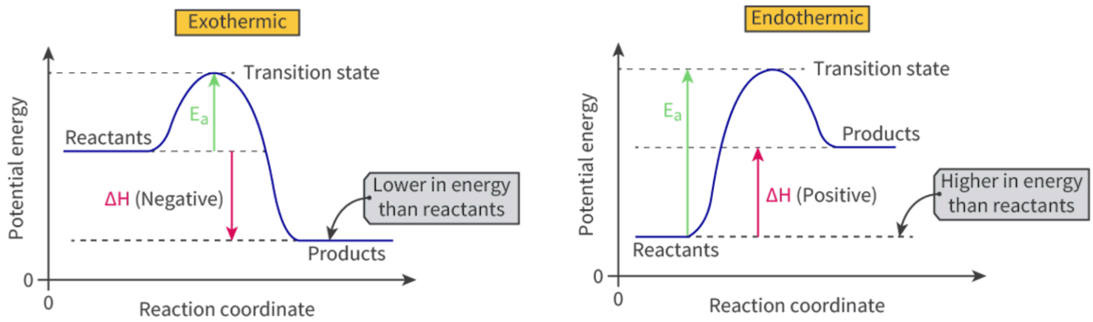

Figure 4. Energy profile diagrams for exothermic and endothermic reactions.

More information for figure 4

The image contains two energy profile diagrams illustrating exothermic and endothermic reactions. Each graph plots potential energy against reaction coordinate.

1. Exothermic Reaction Diagram (Left):
2. The X-axis represents the reaction coordinate, and the Y-axis represents potential energy.
3. The curve starts at the reactants' energy level, rises to a peak identified as the transition state, and then falls to a lower energy level for the products.
4. The activation energy (E_a) is shown as a green upward arrow from reactants to the peak of the transition state.
5. The change in enthalpy (ΔH) is negative, indicated by a red downward arrow from reactants to products.
6. A label notes that products are lower in energy than reactants.

7. Endothermic Reaction Diagram (Right):

8. Similarly, the X-axis is the reaction coordinate, and the Y-axis shows potential energy.
9. The curve begins at the reactants' energy level, rises to a transition state, and then decreases to a higher energy level for the products.
10. The activation energy (E_a) is depicted as a green upward arrow from the reactants to the peak.
11. The change in enthalpy (ΔH) is positive, marked by a red upward arrow from reactants to products.

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12. A label indicates that products are higher in energy than reactants.

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Energy profile diagrams for a catalysed and uncatalysed reactions

Figure 5 shows the profile diagram for an exothermic and endothermic reaction with and without a catalyst.

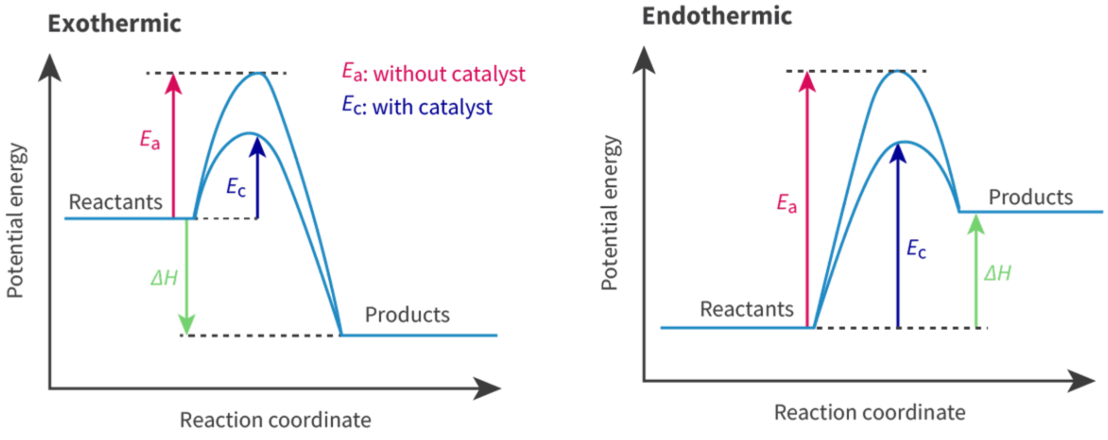

Figure 5. An exothermic and endothermic reaction profile diagram showing a catalysed and uncatalysed reaction.

More information for figure 5

The diagram illustrates two reaction profiles side by side: one for an exothermic reaction and one for an endothermic reaction. Each profile compares catalyzed and uncatalyzed reactions.

On the left is the exothermic reaction graph. The X-axis is labeled 'Reaction coordinate,' and the Y-axis is labeled 'Potential energy.' The graph starts with 'Reactants' at a relatively high energy level, peaking at the transition state. The peak labeled as 'E_a' indicates the activation energy without a catalyst. A lower peak labeled 'E_c' shows the reduced activation energy with a catalyst. The graph descends as it moves to 'Products,' showing a decrease in potential energy, marked by 'ΔH' indicating an overall energy release.

On the right is the endothermic reaction graph. It has similar axis labels to the exothermic graph. The 'Reactants' start at a lower potential energy. There are two peaks, with 'E_a' showing the higher activation energy without a catalyst, and 'E_c' displaying the lower activation energy with a catalyst. The potential energy increases in the products, highlighted by 'ΔH' indicating energy absorption.

Overall, both diagrams illustrate how catalysts lower the activation energy needed for reactions, affecting energy absorption and release differently in exothermic and endothermic processes.

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The catalyst provides an alternative route with lower activation energy; the profile diagrams in **Figure 5** show this. The activation energy for the catalysed reaction is much lower than the activation energy of the uncatalysed reaction.

Study skills

- Note that the activation energy for the endothermic reaction spans the entire height of the peak and not just the hump.
- Remember that ΔH remains the same regardless of whether a catalyst is used or not. The $E_{cat} < E_a$ hence the rate of reaction is faster in both the catalysed exothermic and endothermic reactions

Maxwell-Boltzmann distribution curves

Temperature is a measure of the average kinetic energy of the molecules. In a sample at a given temperature, not all particles will possess the same amount of energy. This means that at any given temperature some fraction of the molecules will possess very high energy and will move very fast. Another fraction will have very low energy and will move very slowly. But most molecules have energy between these two extremes. The Maxwell-Boltzmann distribution curve shows the distribution of these energies (**Figure 6**). The *y*-axis on the Maxwell-Boltzmann graph can be thought of as giving the number of molecules possessing the given energy. This is very similar to say a distribution of heights graph for the pupils in your classroom. If you plot the range of heights on the (*x*-axis) and the number of students on the *y*-axis, the *y*-axis will give you the number of students falling in a specific height range. In the same way the *x*-axis on the Boltzmann curve gives the amount of energy that the particles possess.

The shape of the Maxwell-Boltzmann distribution curve is not a bell curve, showing that the distribution of energies is not a normal distribution. In fact, it is shifted to the left, showing that there are more particles with lower energy and fewer with higher energy.

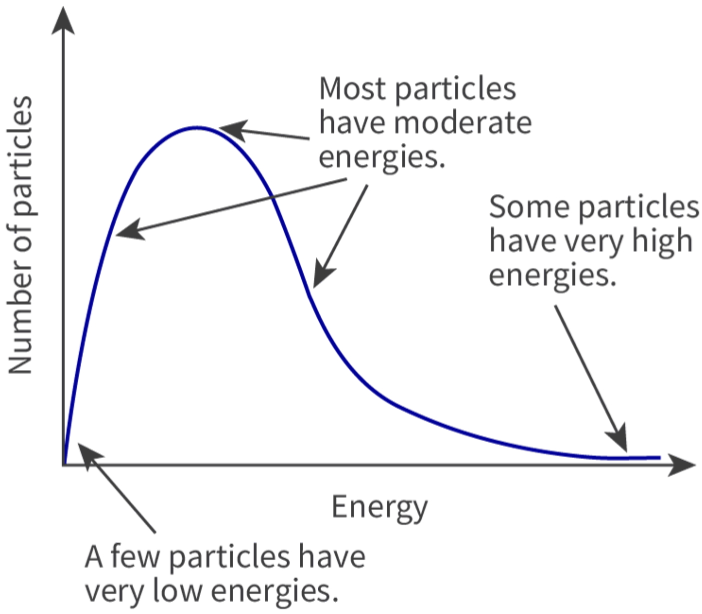

Figure 6. The Maxwell-Boltzmann distribution curve — energy distribution in the molecules of a gas.

Note the area under the curve is a measure of the total number of particles present.

More information for figure 6

This image is a graph depicting the Maxwell-Boltzmann distribution curve for energy distribution among gas molecules. The X-axis represents 'Energy' and the Y-axis represents 'Number of particles'. The curve illustrates that most particles have moderate energies, peaking in the middle of the graph. There are fewer particles with very high energies towards the right, and fewer with very low energies towards the left. The graph highlights key areas: the left shows "A few particles have very low energies", the peak shows "Most particles have moderate energies", and the right shows "Some particles have very high energies". The curve is not symmetric, showing a skewed distribution to the left, indicating a higher number of particles with lower energy.

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The Maxwell-Boltzmann distribution curve and activation energy

When particles collide, they must have energy equal to or greater than the activation energy in order for the reaction to occur. The activation energy can be marked on the Maxwell-Boltzmann curve as shown in **Figure 7**.

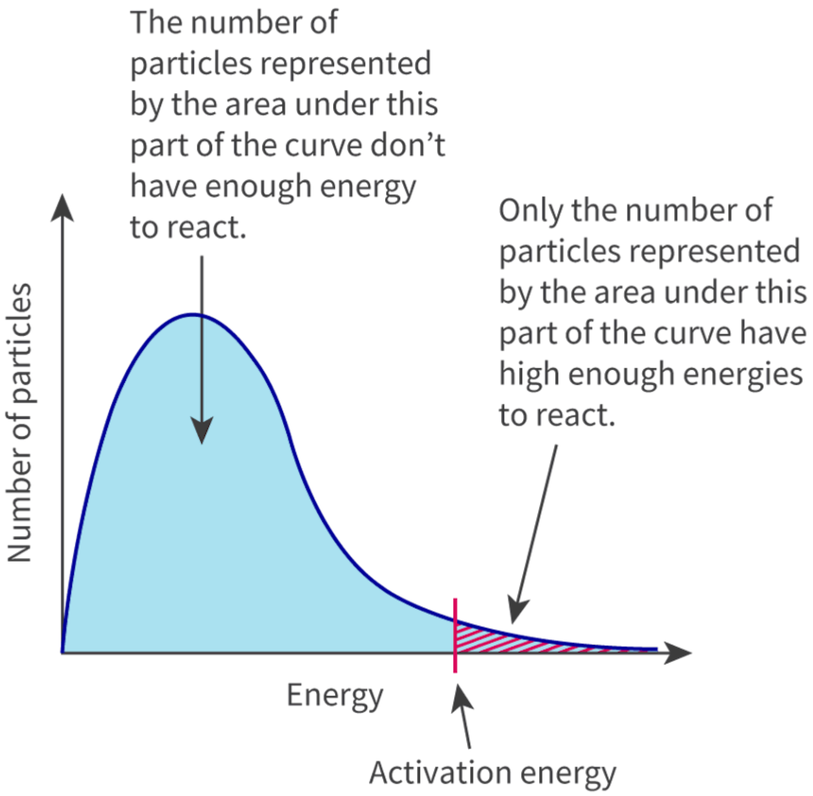

Figure 7. The Maxwell-Boltzmann distribution curve and activation energy.

More information for figure 7

The image is a graph illustrating the Maxwell-Boltzmann distribution curve related to the kinetic energy of particles. The X-axis is labeled as "Energy," representing the kinetic energy of particles, while the Y-axis is labeled as "Number of particles." The graph depicts a bell-shaped curve that peaks and then declines.

To the left of the curve's peak, an annotation indicates that the number of particles under this part of the curve does not have enough energy to react. To the right of the curve, another annotation points out that only the number of particles under this part of the curve has high enough energies to react.

A vertical line is drawn from the X-axis, intersecting the curve, marking the "Activation energy," which is the minimum energy required for the particles to react upon collision. The area under the curve to the right of this line represents particles with sufficient energy for a reaction, while the area to the left represents particles lacking this energy.

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In **Figure 7**, the area under the curve to the right of the activation energy line is the number of particles that have the necessary energy to react on collision. Particles on the left side of the activation energy line do not possess the minimum energy required for a successful collision.

Study skills

Remember!

- The *x*-axis on the curve is labelled 'Energy' and the *y*-axis is labelled as 'Number of particles'
- The curve must always start at zero.
- The curve is not symmetrical and tapers off to the right to form a 'tail'.

The Maxwell-Boltzmann curve and the effect of temperature

An increase in temperature increases the reaction rate. This is because there is an increase in the average kinetic energy of the reactant particles and the collision frequency increases. This means that a greater fraction of the particles possess energy equal to or greater than the activation energy. **Figure 8** shows the distribution of kinetic energy at two different temperatures.

Note that the curve for higher temperature has a lower peak height, this is because it has been slightly stretched to the right. You can now see that more particles now possess higher energy for this curve, that is more particles now have energy greater than the activation energy. Although the shape of the two curves is different, the area under the curves remains the same, that is the number of particles is unchanged.

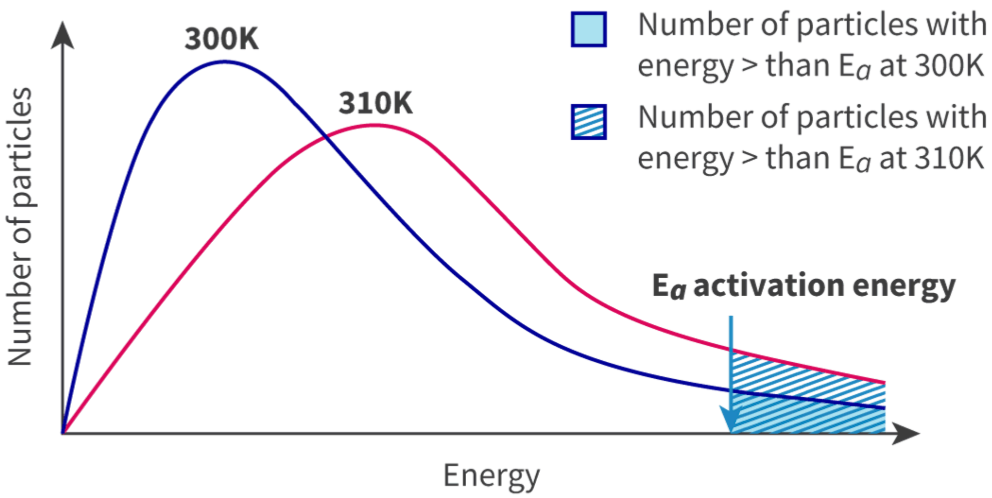

Figure 8. The Maxwell-Boltzmann distribution curve at two temperatures.

More information for figure 8

The image depicts two Maxwell-Boltzmann distribution curves on a graph. The X-axis represents energy, while the Y-axis represents the number of particles. The two curves illustrate the distribution of particle energy at different temperatures: one for 300K and another for 310K. The 300K curve is colored blue, while the 310K curve is pink. Both curves start from the origin, rise to a peak, and then fall asymptotically back towards the X-axis. The 310K curve has a lower peak height, indicative of increased energy distribution, and is shifted to the right compared to the 300K curve. There are shaded areas under each curve to the right of an indicated activation energy (Ea) line, with the area under the 310K curve having more particles with energy greater than Ea, represented by striped shading.

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The shaded area under each curve to the right of the activation energy is the fraction of particles that have sufficient energy to react on collision. The striped area under the higher temperature graph is larger than the blue area under the lower temperature curve. Therefore, at higher temperatures, a greater proportion of the collisions will be successful.

Maxwell-Boltzmann distribution curves and the effect of catalysts

The way that a catalyst speeds up a reaction can be explained using a Maxwell-Boltzmann curve. The presence of the catalyst does not alter the average kinetic energy of the molecules but provides an alternate route with lower activation energy.

This shifts the activation energy line to the left, so that there are now more particles meeting the E_a requirement (**Figure 9**).

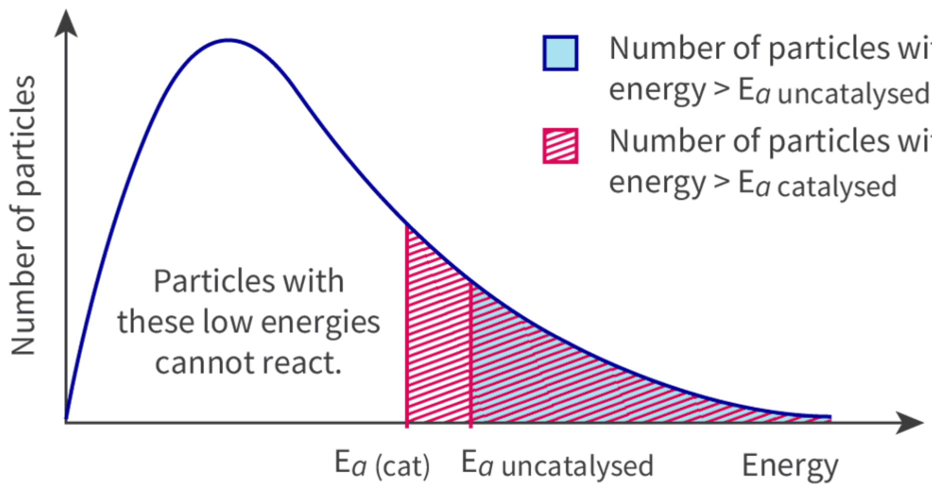

Figure 9. The Maxwell-Boltzmann diagram for a reaction with and without a catalyst.

More information for figure 9

The diagram is a Maxwell-Boltzmann distribution graph. The X-axis represents energy, while the Y-axis represents the number of particles. There are two activation energy lines marked: Ea (cat) and Ea uncatalysed. To the right of these lines are shaded areas under the curve, illustrating the number of particles with sufficient energy to react. The area to the right of the Ea (cat) is larger, indicating more particles meet the catalysed energy requirement compared to the uncatalysed one. A section in blue shows the number of particles with energy greater than Ea uncatalysed, and a section in red with stripes shows particles with energy greater than Ea catalysed. A note on the left indicates particles with low energies cannot react.

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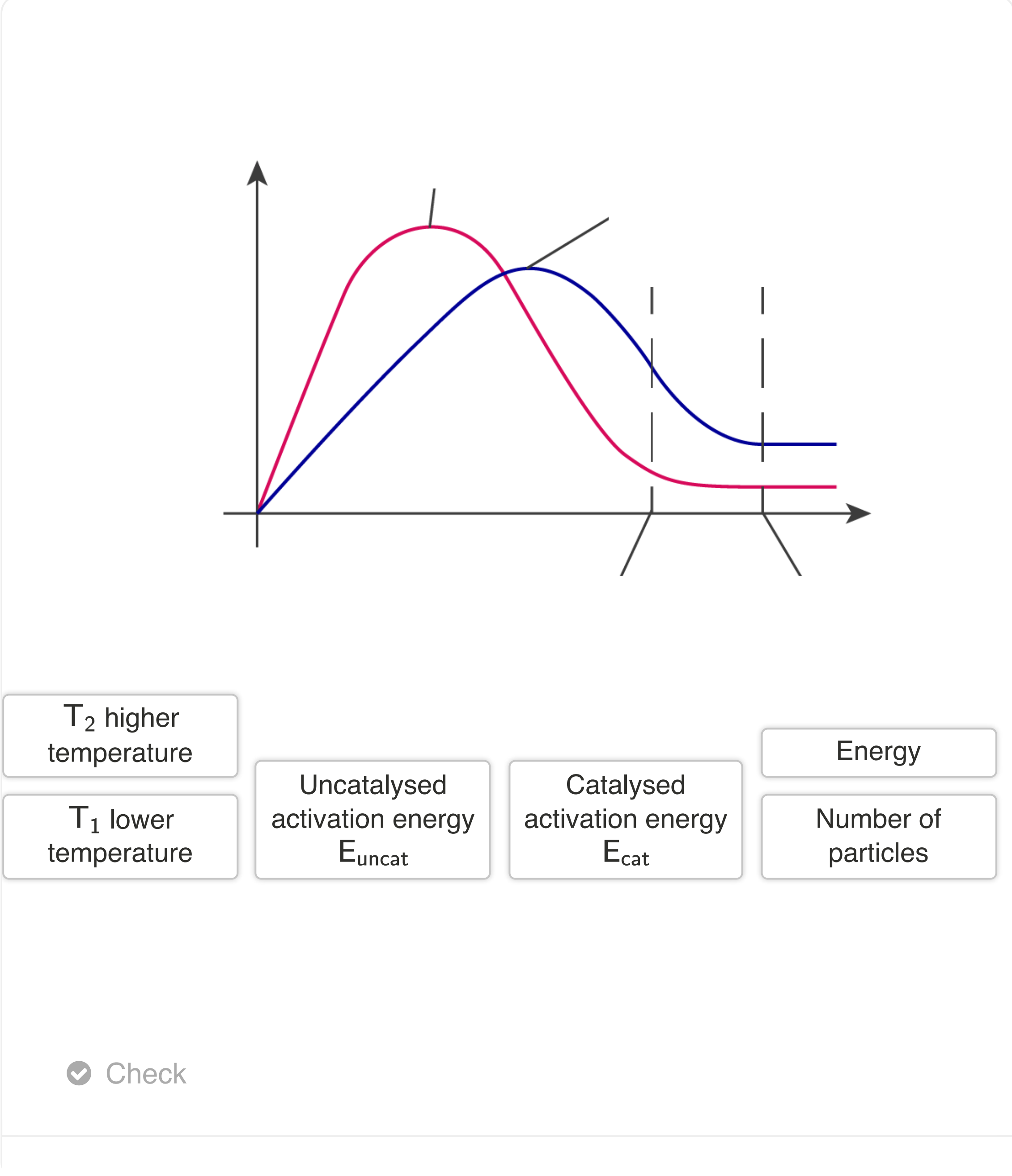
The shaded area on the right of each activation energy line is the number of particles that have sufficient energy to react upon collision. Activation energy of the catalysed reaction, $E_{a\ (cat)}$, is lower than the uncatalysed activation energy, $E_{a\ (uncat)}$, so the striped area under the curve to the right of the $E_{a\ (cat)}$ is greater than the area under the curve to the right of the $E_{a\ (uncat)}$. This shows that a greater proportion of the collisions involving the catalyst will be successful. Therefore, the rate of reaction will be greater in the presence of the catalyst.

Use **Interactive 1** to drag and drop the labels to the correct places.



Interactive 1. The Maxwell-Boltzmann Distribution Curve — Energy Distribution in the Molecules of a Gas.

More information for interactive 1

This interactive Maxwell—Boltzmann distribution graph illustrates how temperature and the presence of a catalyst influence the energy distribution of reacting particles in a system.

- The graph shows two bell-shaped curves, both skewed to the right.
    - Red curve represents particles at a lower temperature (T_1).
    - The blue curve represents particles at a higher temperature (T_2).
- The Y-axis indicates the number of particles.
- The X-axis shows the energy of the particles.
- Two dashed vertical lines near the tail represent:
    - The activation energy without a catalyst (E_{uncat}).
    - The activation energy with a catalyst (E_{cat}), which is lower.

Users must drag and drop the correct labels into the appropriate blank zones on the graph, including:

- The X-axis and Y-axis
- The red and blue curves
- The two activation energy lines near the curve tails

The drag-and-drop options are as follows:
T_2 higher temperature
T_1 lower temperature
Uncatalysed activation energy c
Catalysed activation energy E_{cat}
Energy
Number of particles
This interactivity helps users understand how temperature and catalysts affect the distribution of particle energies.
Read below for the solution:
Y-axis - Number of particles
X-axis - Energy
Red curve - T_1 lower temperature
Blue curve - T_2 higher temperature
First section - Catalysed activation energy E_{cat}
Second section - Uncatalysed activation energy E_{uncat}
This activity reinforces that: At **higher temperatures**, a **greater proportion of particles** have enough energy to overcome the activation barrier.
Catalysts lower the activation energy, thereby **increasing the number of particles** that can react.

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Inquirers
  - Thinkers
  - Reflective
- **Approaches to learning:**
  - Thinking skills — Providing a reasoned argument to support conclusions; Applying key ideas and facts in new situations
  - Communication skills — Use terminology, symbols and communication conventions consistently and correctly
  - Social skills — Working collaboratively to achieve common goals; Appreciating the talents and diverse needs of others
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual/pair activity

1. A student wrote the following explanation for how a catalyst works:

   'Sulfur dioxide and oxygen react to give sulfur trioxide, in the presence of a vanadium pentoxide catalyst. For the reaction to occur the sulfur dioxide and oxygen molecules must collide. The catalyst lowers the activation energy of this collision.'

   Identify the errors and suggest a better answer.
2. Define the term activation energy of a reaction.
3. (a) Draw, with labelled axes, the Maxwell-Boltzmann curve to show the distribution of molecular energies in a gas. Label this curve T_1. Draw a second curve T_2 to show the same sample of gas at a higher temperature.
   (b) Draw two vertical lines showing the activation energy on the Maxwell-Boltzman curve for the uncatalysed, $E_{a\ (uncat)}$, and catalysed, $E_{a\ (cat)}$ reactions.
   (c) Explain why altering the temperature alters the frequency of successful collisions.
   (d) What does the area under the curve T_1 represent and how is it different from the area under the curve T_2.
4. Mark the activation energy of the backward reaction on the following energy profile diagram (**Figure 10**). Sketch what the curve would look like in the presence of a catalyst.

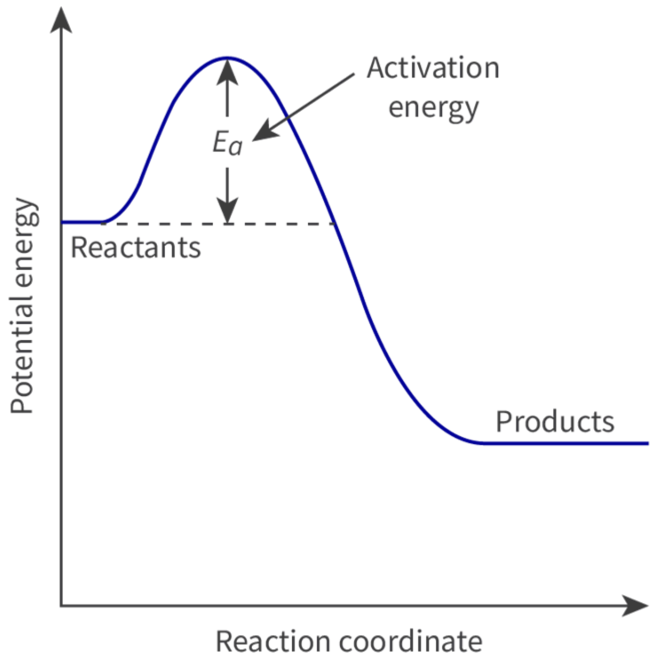


Figure 10. Energy profile diagram.

More information for figure 10

The diagram is an energy profile graph showing the relationship between potential energy and reaction coordinate. The X-axis represents the reaction coordinate, and the Y-axis represents potential energy. The graph features a curve that starts at a certain energy level, increases to a peak, and then decreases. This increase represents the activation energy (labeled as 'Ea'), which is the energy barrier that must be overcome for the reaction to progress from reactants to products. The reactants are labeled at the initial low-energy level prior to the increase, while the products are labeled at the lower energy level after the peak. Arrows indicate the direction from reactants to products along the reaction coordinate, and a dotted line depicts the energy level of reactants beneath the peak, highlighting the activation energy required for the reaction. The overall shape of the graph indicates an exothermic reaction, with products being lower in energy than reactants.

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1. The student identifies that the reaction is catalysed by vanadium pentoxide but it is not about the collisions between the molecules. The catalyst does not lower the activation energy but provides an alternative route through which the reaction can proceed. It is this route that has a lower activation energy.
2. The activation energy, E_a, is the minimum kinetic energy that the colliding molecules must possess for the collisions to be successful and result in the formation of a product molecule.
3. (a) and (b)

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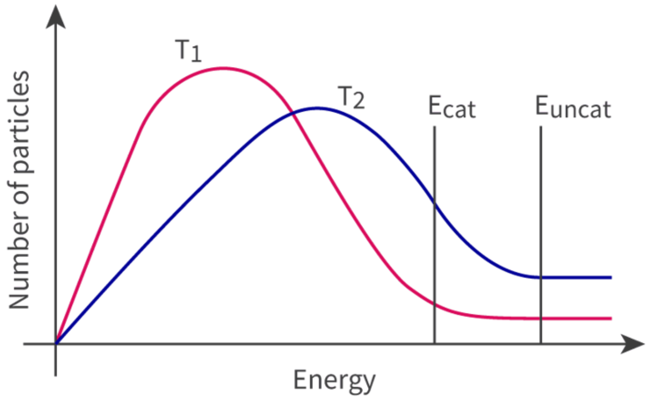

1. (c) Temperature is a measure of the average kinetic energy of the molecules. Increasing the temperature, for example, leads to an increase in kinetic energy, this would lead to an increase in the frequency of collisions.
   (d) The area under curve T_1 represents the total number of particles in the sample. The shape of the curve changes in T_2, but the area under the curve still remains the same, as long as the number of particles is not altered.
2. The catalysed profile and activation energy of the reverse reaction: